IAS PRELIMS/MAINS

Ecology & Environment

A COMPREHENSIVE COVERAGE

MODULE - 14

AIR POLLUTION - 1

AIR POLLUTION

Air pollution may be defined as the presence of any solid, liquid or gaseous substance including noise and radioactive substances in the atmosphere in such concentration that is injurious to humans or other living organisms, or interferes with the normal environmental processes.

The Air (Prevention and Control of Pollution) Act 1981 was amended in 1987 to add 'Noise' to the definition of air pollution.

Learning Space

CLASSIFICATION OF AIR POLLUTANTS

Classification based on the form in which they persist in the environment after their release.

- **Primary pollutants**: It is a pollutant which is directly emitted from a source.
  - ✓ E.g.– SO_2 and NO_x are directly emitted from coal based thermal power plants.
- **Secondary Pollutants**: These are formed by interaction among the primary pollutants.
  - ✓ For example, peroxyacetyl nitrate (PAN) is formed by the interaction of nitrogen oxides and hydrocarbons. Or formed by the interaction between primary pollutants and the atmospheric gases.

CLASSIFICATION OF AIR POLLUTANTS

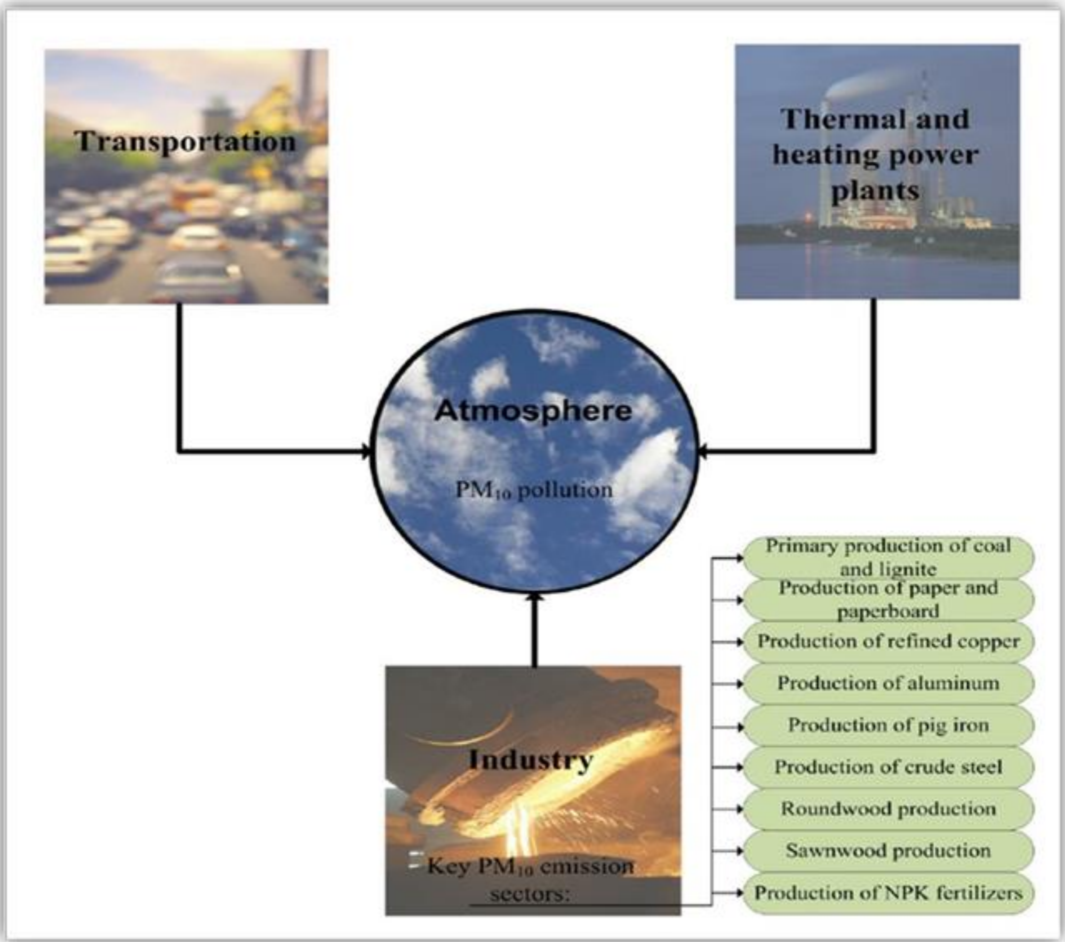
Classification according to their existence in nature

- **Quantitative Pollutants**: These occur in nature and become pollutant when their concentration reaches beyond a threshold level.
  E.g: carbon dioxide, nitrogen oxide.
- **Qualitative Pollutants**: These do not occur in nature and are man-made.
  E.g: fungicides, herbicides, DDT etc.

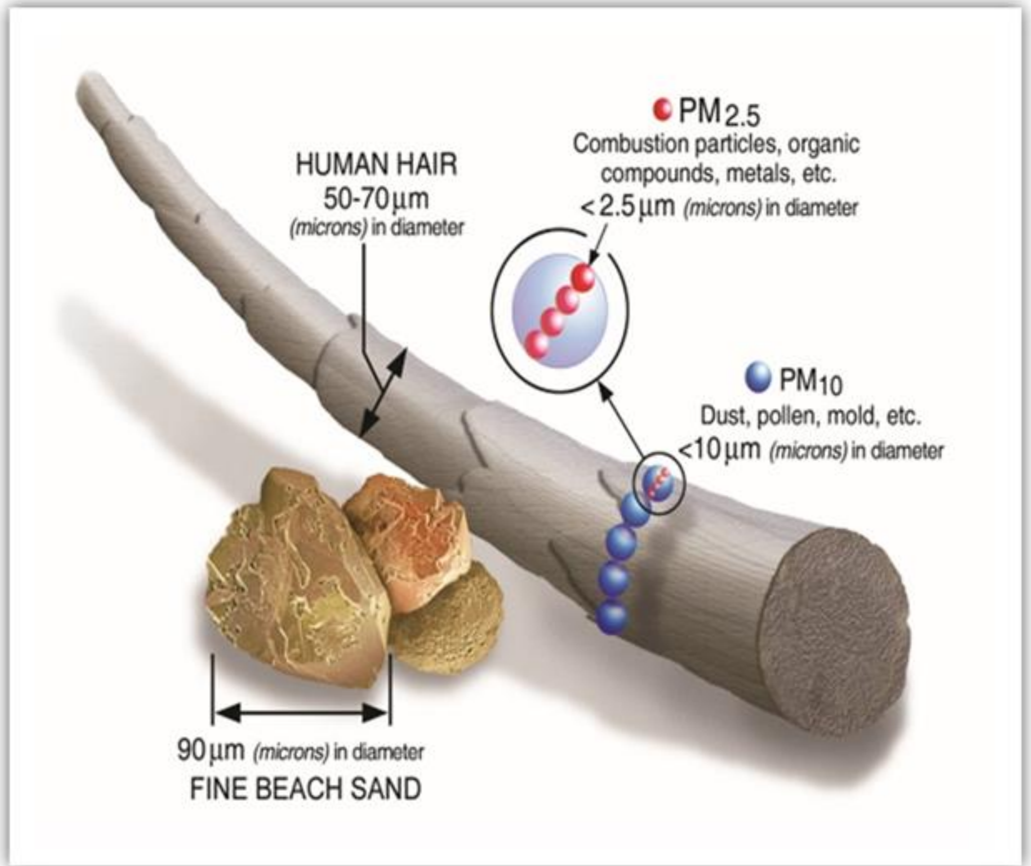
Learning Space

PARTICULATE MATTER

- Particulate matter is released from the industrial chimneys, automobiles, burning of biomass, coal based thermal power plants etc.
- Their size ranges from 0.001 to 500 micrometres (μm) in diameter.
- Particles less than 10 μm float and move freely in the air. Particles which are more than 10 μm in diameter settle down on the ground.



PARTICULATE MATTER



COMMON PARTICULATE MATTER (PM) SIZES

PM_{10}

All particles with aerodynamic diameter <10 micro meter

$PM_{2.5}$

All particles with aerodynamic diameter <2.5 micro meter

PM_1

All particles with aerodynamic diameter <1 micro meter

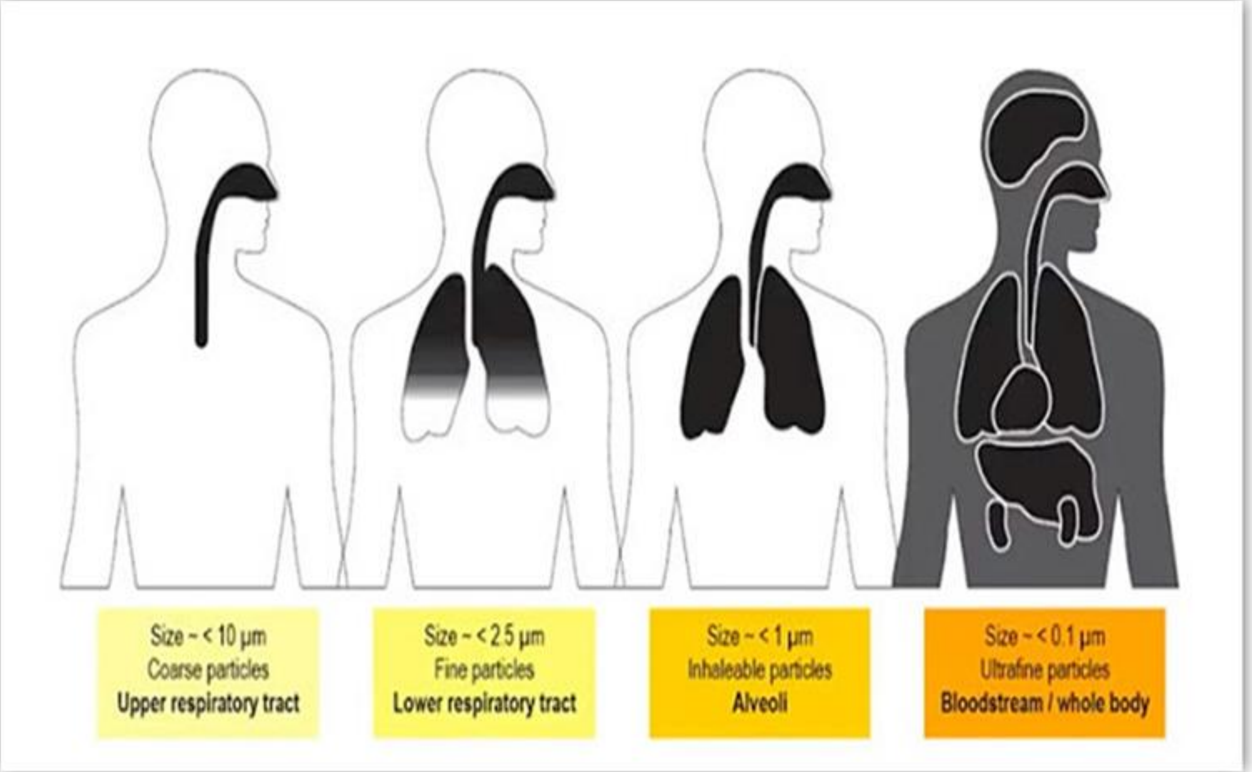
UFP

All particles with aerodynamic diameter <100 micro meter

PARTICULATE MATTER

Health effects of PM 2.5

- These particles are able to travel deep into the respiratory tract, reaching the lungs. Exposure to these particles causes
  - ✓ Throat irritation
  - ✓ Coughing
  - ✓ Sneezing
  - ✓ Shortness of breath etc.
- Nitrogen particles make up the largest fraction of PM 2.5.



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